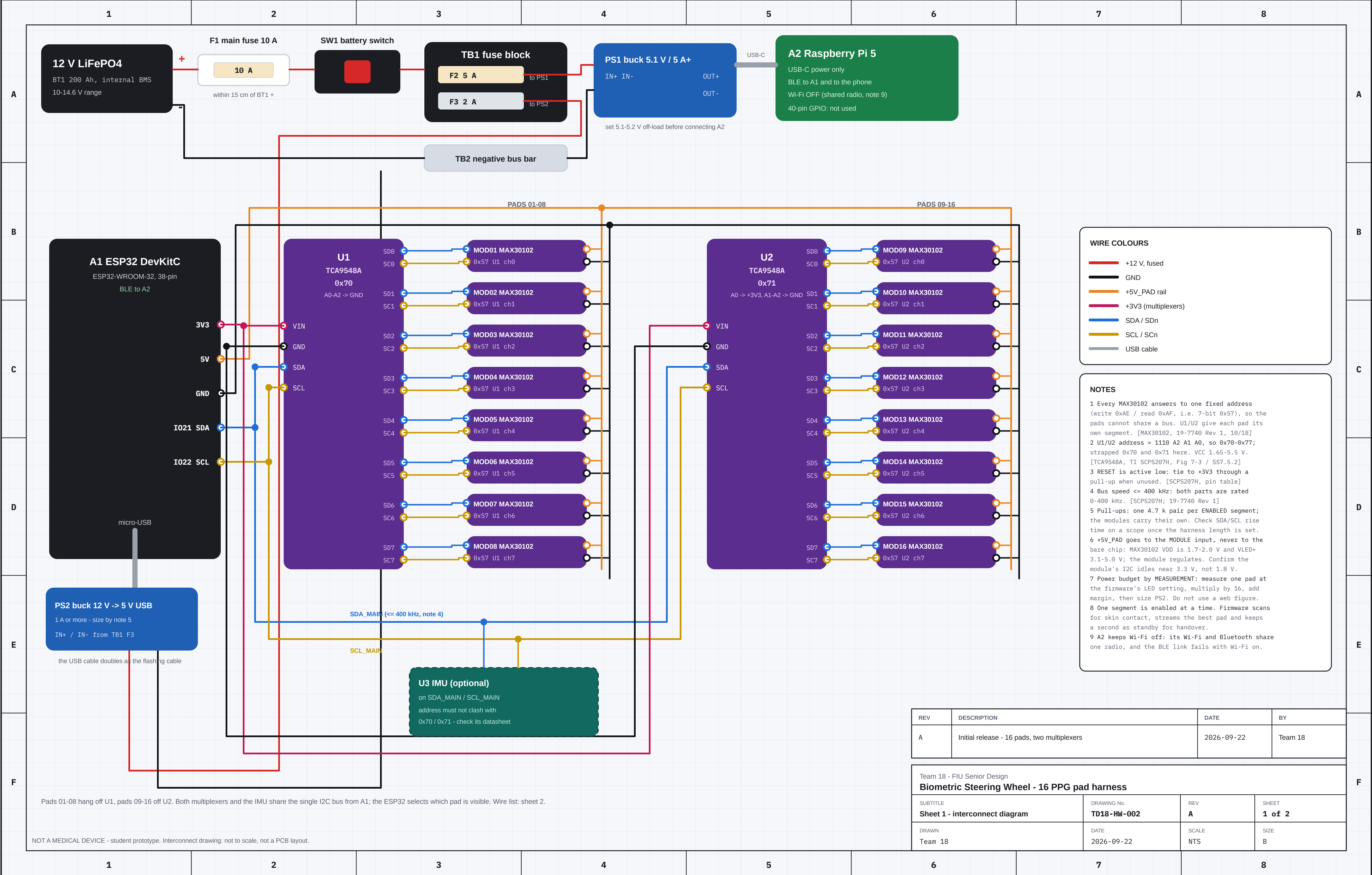




	1	2	3	4	5	6	7	8	
A	PARTS LIST								
	REF	QTY	DESCRIPTION	KEY RATINGS / NOTES					
	BT1	1	12 V LiFePO4 battery, 200 Ah, internal BMS	10-14.6 V working range					
	F1	1	Fuse + holder, 10 A	within 15 cm of BT1 +					
	SW1	1	Battery master switch	disconnects the whole system					
	TB1	1	Fused distribution block, 2 ways	F2 5 A -> PS1, F3 2 A -> PS2					
	TB2	1	Negative bus bar	single return point					
	PS1	1	Buck converter 12 V -> 5.1 V, 5 A or more	set off-load before connecting A2					
	PS2	1	Buck converter 12 V -> 5 V USB	1 A or more; size by note 5, sheet 1					
	A1	1	ESP32-WROOM-32 DevKitC, 38-pin	I2C on IO21/IO22; BLE to A2					
	A2	1	Raspberry Pi 5	USB-C power only; Wi-Fi off					
	U1, U2	2	TCA9548A 8-channel I2C switch module	0x70 / 0x71; VCC 1.65-5.5 V; 400 kHz					
	MOD01-16	16	MAX30102 pulse-oximetry module	0x57; module VIN, onboard regulator					
	U3	1	IMU module (optional)	on the main bus; address must not clash					
	R1-R4	a/r	Pull-up resistors 4.7 k	one pair per enabled segment (note 5)					
R5, R6	2	Pull-up resistors 10 k	RESET of U1/U2 to +3V3 (note 3)						
-	a/r	Harness wire, sleeving, strain relief	size from the measured current (note 7)						
B									
	C	WIRE LIST							
FROM		TO		NET		WIRE			
BT1 +		F1 in		+12V_RAW		red			
F1 out		SW1 1		+12V_FUSED		red			
SW1 2		TB1 in		+12V_SW		red			
TB1 F2 out		PS1 IN+		+12V_PI		red			
TB1 F3 out		PS2 IN+		+12V_ESP		red			
BT1 -		TB2		GND		black			
TB2		PS1 IN- , PS2 IN-		GND		black			
PS1 OUT+ / OUT-		A2 USB-C		+5V_PI		USB cable			
PS2 OUT		A1 micro-USB		+5V_ESP		USB cable			
A1 3V3		U1 VIN, U2 VIN		+3V3		pink			
A1 5V		MOD01-16 VIN		+5V_PAD		orange			
A1 GND		U1/U2 GND, MOD01-16 GND, U3 GND		GND		black			
A1 IO21		U1 SDA, U2 SDA, U3 SDA		SDA_MAIN		blue			
A1 IO22	U1 SCL, U2 SCL, U3 SCL		SCL_MAIN		yellow				
D	U1 SD0..SD7	MOD01..MOD08 SDA		SDA_A0..A7		blue			
	U1 SC0..SC7	MOD01..MOD08 SCL		SCL_A0..A7		yellow			
	U2 SD0..SD7	MOD09..MOD16 SDA		SDA_B0..B7		blue			
	U2 SC0..SC7	MOD09..MOD16 SCL		SCL_B0..B7		yellow			
	U1 A0, A1, A2	GND		ADDR_U1 = 0x70		-			
	U2 A0	+3V3 (A1, A2 to GND)		ADDR_U2 = 0x71		-			
	U1 RESET, U2 RESET	+3V3 through 10 k		RST_MUX		-			
	E								
F									
	NOT A MEDICAL DEVICE - student prototype. Interconnect drawing: not to scale, not a PCB layout.								
1	2	3	4	5	6	7	8		

Team 18 - FIU Senior Design			
Biometric Steering Wheel - 16 PPG pad harness			
SUBTITLE Sheet 2 - parts list and wire list	DRAWING No. TD18-HW-002	REV A	SHEET 2 of 2
DRAWN Team 18	DATE 2026-09-22	SCALE NTS	SIZE B